# Bargaining over Leasing Contracts: Strong by Privilege but Weak by Risk Aversion

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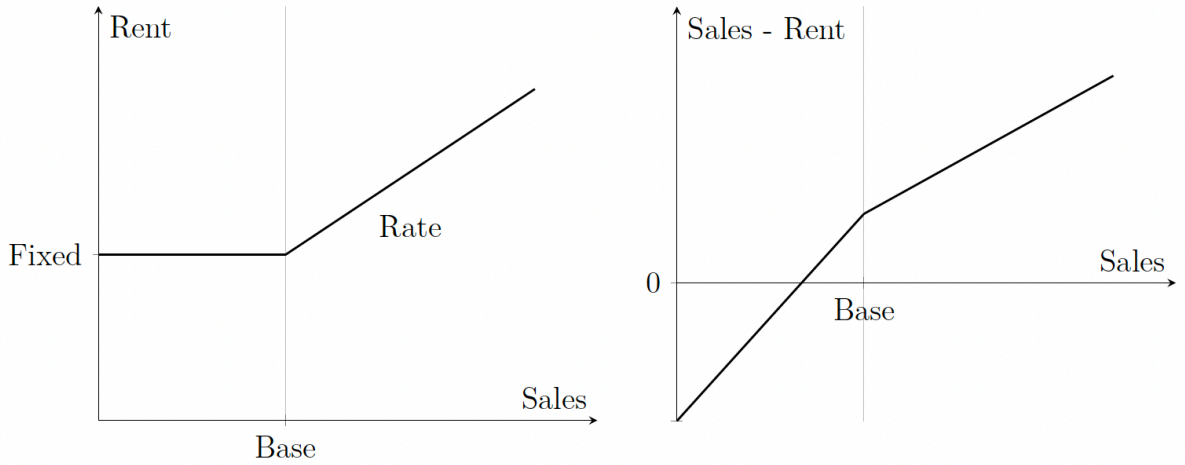
# (Un)fair Trading in Complex Transactions

- Fair trading benefits the economy, and regulators like the FTC enforce it
- How can they detect unfair practices? $\rightarrow$ Clear evidence is required
  - Recordings of collusive meetings / High markups
- Problem: Real business transactions involve complex transfer schemes
- Example: Tenant leasing in shopping mall




(a) Share of Shopping Mall (b) Share of Tenant

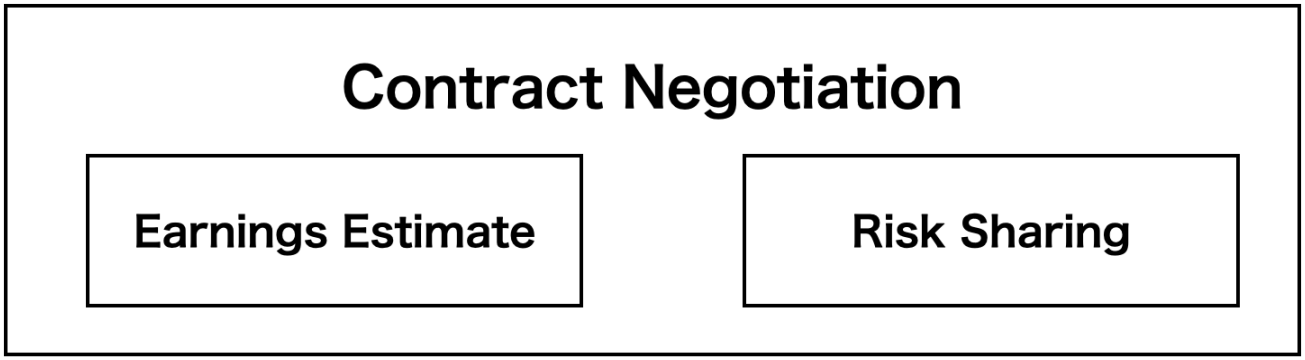
Figure 1. How Sales is Shared with Shopping Mall and Tenant

- Question: How does power balance appear in contract form?

# Data Source

- Data covers two shopping malls over 6 years (2017-2023)
  - Managed by same company, by different managers
  - Located in same region (Western region of Japan)
  - One in downtown (*Mall 0*), one in suburb (*Mall 1*)
- For them, we have the following two data sources:
  1. performance data
  2. contract data
- They are used for actual leasing operation of the management company
- I focus on contract renewals not contracts with new entrants
  - In total, 226 tenants operate under 443 contracts

# Background of Contract Negotiation

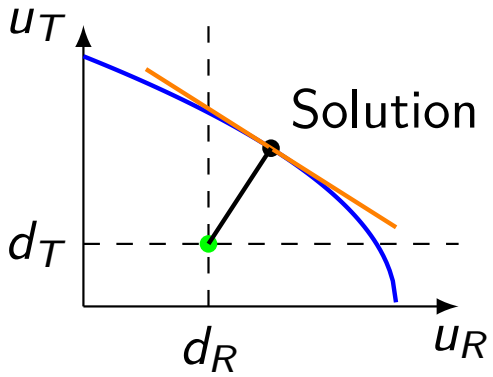



- Mall and Tenant has a negotiation table for a new contract term
- Each contract negotiation is composed of two tasks
  - **Earnings estimate**: Derive the sales distribution for the next period
  - **Risk sharing**: Shift the risk involved in the estimate toward each other
- Earnings estimate is not a simple observed average sales:
  - Tenant tries to increase the earnings estimate
  - Mall sticks to the realized sales history

# Model of Contract Negotiation

**Bargaining for Earnings Estimate** → **Bargaining for Risk Sharing**

- The two tasks are sequentially resolved through separate Nash bargaining
- Solution concept: **Nash-in-Nash solution** by Horn and Wolinsky 1988
  - Multiple interrelated Nash bargaining problems are solved simultaneously under "other problems fall into the Nash bargaining solution" assumption
- Reminder: **Nash bargaining solution**
  - Primitives:
    - Bargaining frontier
    - Bargaining power ratio
    - Break-up point
  - Outcome: Surplus split between parties




- Bargaining power ratio in the first table is considered as *privilege*

# Result: Privilege → Risk Averse → Favor Fixed Rent

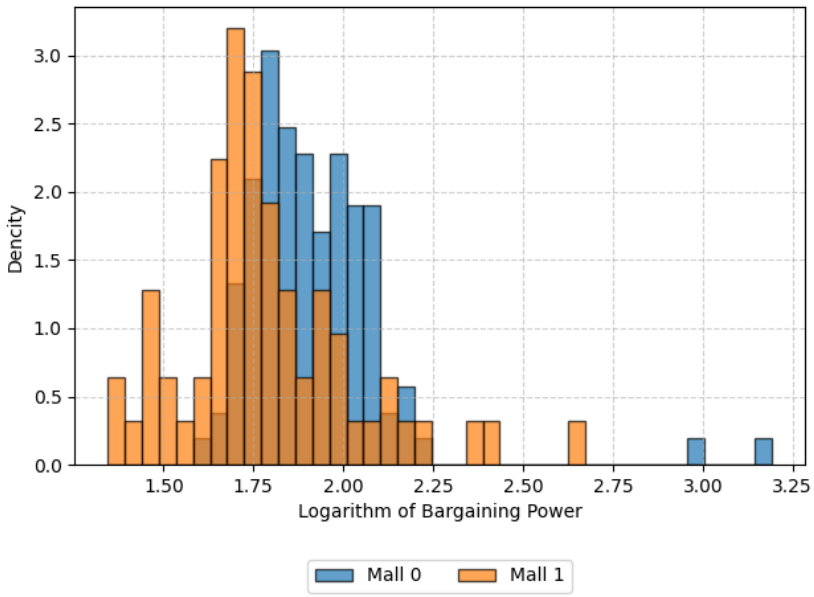



Figure: *Privilege*

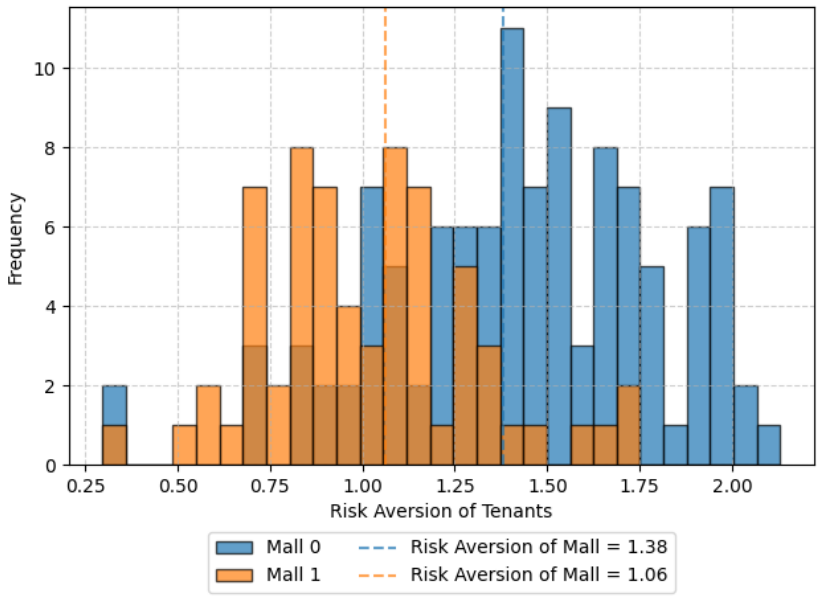



Figure: Risk aversions

- Mall 0 is more privileged due to high-traffic area
  → Earnings estimate is set to lower value
  → Commission component does not yield much rent
  → Mall manager emphasizes Fixed, i.e., she becomes more risk averse

# Result: Privilege → Forego Fixed Rent

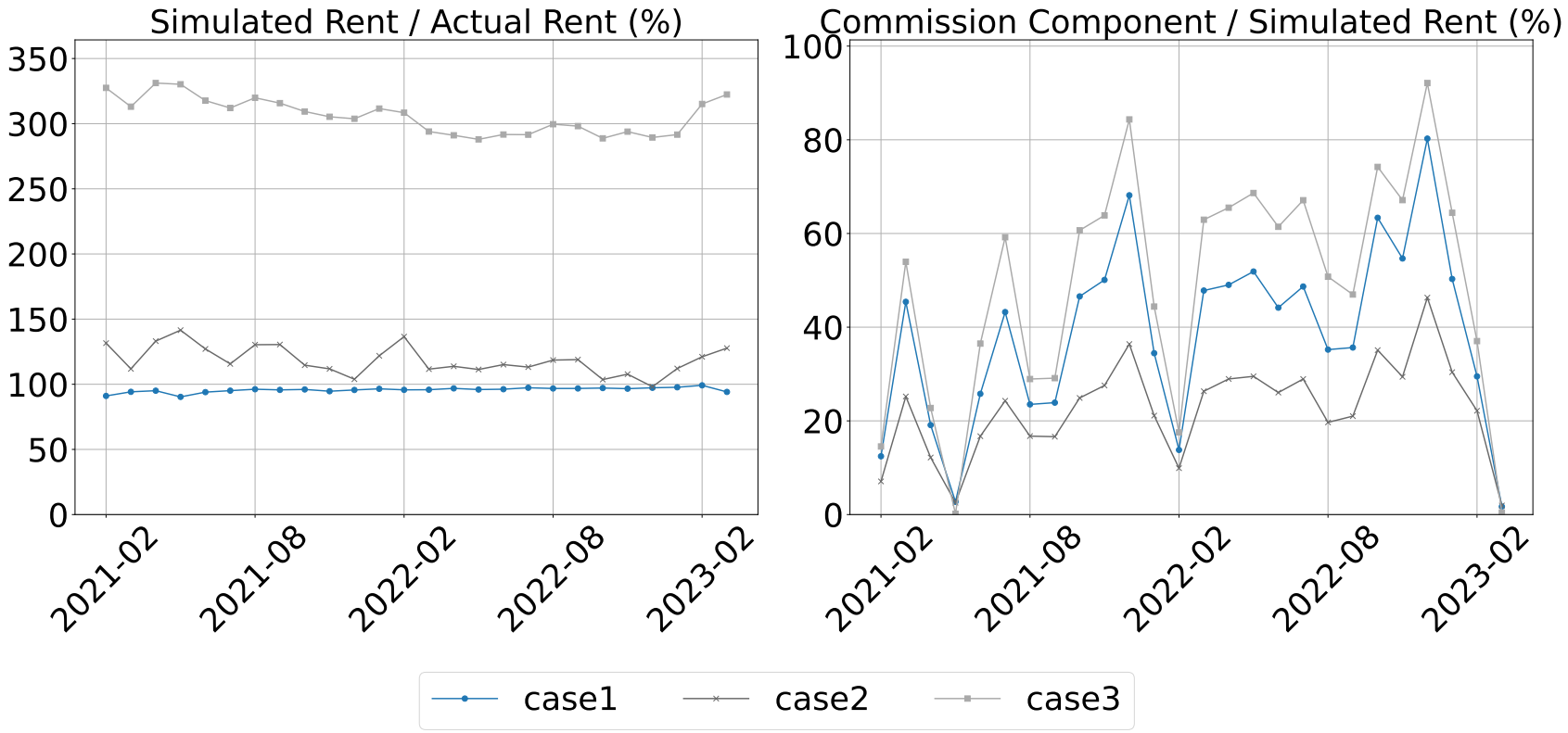
| Mall | Mall 0 | | | Mall 1 | | |
|---|---|---|---|---|---|---|
| Dep. Var | Rate | Base | Fixed | Rate | Base | Fixed |
| Avg. Sales | 0.00204*** | -0.135 | 0.00922 | 0.00533*** | 0.461*** | 0.0274 |
| | (0.000720) | (0.0843) | (0.0134) | (0.000962) | (0.121) | (0.0202) |
| Var. Sales | 0.000881 | 0.671*** | 0.0552** | -0.00307 | -0.130 | 0.0299 |
| | (0.00116) | (0.136) | (0.0216) | (0.00321) | (0.404) | (0.0675) |
| Privilege | -0.0480*** | -2.987* | -0.0569 | -0.0400*** | -3.086** | -0.874*** |
| | (0.0136) | (1.590) | (0.253) | (0.0101) | (1.275) | (0.213) |
| Risk loving | 0.125*** | -2.154 | 0.341 | 0.0985*** | -1.350 | -0.157 |
| | (0.0121) | (1.416) | (0.225) | (0.00789) | (0.994) | (0.166) |
| *N* | 90 | 90 | 90 | 62 | 62 | 62 |

- Findings:
  - Privilege leads to smaller fixed and lower base
    - Why? Privileged mall is more pessimistic about earnings estimate
  - Contract selection rule itself is consistent with risk attitude
    - Mall 0 (Risk averse): Large volatility → Higher Fixed
    - Mall 1 (Risk loving): Higher average sales → Higher base and Higher rate

# Simulation Setting: Fairer Trade

- Situation:
  - Mall 0 potentially abuses its privilege when making contracts
  - Regulator enforces the fairer bargainings in renewal through warnings
- Question: How the amount of rent and its composition change?
  - Why is this an empirical question? $\rightarrow$ Two paths exist
    1. Weaker positions $\rightarrow$ Higher earnings estimate $\rightarrow$ Larger fixed rent
    2. Weaker position $\rightarrow$ Less risk averse $\rightarrow$ More commission component
- Scenarios:
  - Case 1: Replicate actual rents
  - Case 2: Mall 0's privilege is determined in the same way as in Mall 1
  - Case 3: Case 2 + Mall 0's risk aversion is set to the same value of Mall 1

# Result: Fairer bargaining does not always lead to less rent




- Even in Case 2, the fairer situation does not always yield less rent
- In Case 3, I find sharp increase in commission component
- This increase could triple the amount of rent